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Q1. Write a program that uses a while loop to calculate

Q2. Write a program that uses a while loop to find the

- `num = 6`

Q3. Write a program using a while loop that prints num

Q4. Write a while loop to print the first 20 even number

Q5. Write a program that uses a while loop to find the

- `num = 54321`

Q6. Write a program that reverses a number using a w

Q7. Create an infinite while loop that asks the user to e
breaks; otherwise, it prints the square of the number?

Q8. write a loop statement to print the following series

- output : 10,20,30,40,....300
- append them to a list

Q9. write a while loop statement to print the following

- output : 105,98,91.....7.
- append them to list.

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Q16. create an empty list. Continuously ask the user to until they type "done." Use the append() method to add

Q17. write a program to accept 10 numbers from the us

Q18. write a program to print all the characters in strin

Q19. write a python program to print only the odd num

- `list = [23,45,25,32,46,33,71,90]`.

Q20. write a python program to get the following outp

- 1--- 49
- 2--- 48
-
- 49--- 1

Q21. Write a program that uses a while loop to check

- `str = "madam"`

Q22. Write a program using a while loop to print all th

Q23. Write a program to find the least common multip

Q24. Write a program using a while loop to print a rigl



Q1. Write a Python function to find the sum of all elements in a list.

Q2. Write a python program to count the occurrence of each element in a list?

Q3. Write a python code to generate a list of squares for the numbers between 1 to 10?

Q4. Write a Python function that:

- Takes two lists as input.
- Merges the two lists into a single list, alternating elements from each list.
- Returns the merged list.

Q5. Given a list of numbers `numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]`, write a Python program to remove multiple elements from a list?

- to remove 3, 5 and 7

Q6. Given a list of integers `[10, 20, 30, 20, 20, 30, 40, 50, -20, 60, 60, -20, -20]` with duplicate elements in it. Generate another list, which contains only the duplicate elements.

Q7. Given 3 digits 4, 5, and 6. Find all the possible combinations from these digits.

Q8. Write a Python program to create a set of even numbers from the list `[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]`?

Q9. Write a Python program to find the union of two sets `{1, 2, 3}` and `{3, 4, 5}` without using the `union()` method?

Q10. Given a list of integers, write a function that partitions the list into two sets: one for even numbers and one for odd numbers.

Q11. Write a function that groups words that are anagrams of each other into separate sets.

Q12. Write a program to check whether 13 is prime or not?

Q13. Write a program to display sum of odd numbers and even numbers separately that fall between two numbers given from the user?

Q14. Write a Python program to create a set by taking user input for elements?

Q15. Write a program to print the index number of 81 present in the given tuple :

- `t = (1, 4, 9, 16, 25, 36, 49, 64, 81, 100)`

Q16. Write a python program using **for loop** to print 9 table?

Q17. Write a program to print the following pattern :

- 55555
- 4444
- 333
- 22
- 1



Q18. covert the following loop into for loop :

- `x = 4`
- `while (x<=8)`
- `print(x*10)`
- `x+=2`

Q19.Python program to check the validity of password input by users?

Q20. Create a program that prints each element of a list along with its index using a for loop?

Q21. Write a program that prints all numbers between 1 and 50 that are divisible by 3 using a for loop?

Q22. Write a program that accepts a string from user and counts the number of uppercase and lowercase letters using a for loop?

Q23.Write a program that extracts all digits from a given string and prints them as a list?

Q24. Given a list of names, write a program that creates a dictionary where the names are keys and their lengths are values using a for loop?

- `names = ["Alice", "Bob", "Charlie", "David", "Eve"]`

Q25.Accept two strings from user, write a program that finds and prints all common characters (without duplicates) using a for loop?

Q26. Write a program that generates a list of n random integers between 1 and 100 using a for loop?

Q27.Write a Python program to reverse the order of characters in a string without using slicing or any in-built function?

Q28. Write a Python program that counts the occurrences of each character in a given string?

- `s = "for_loop"`

Q29. Write a program that finds the common and unique elements in two lists?

Q30. Write a program to count how many times a specific element appears in a list?

- `l = ["a","b","c","a","a","b"]`
- count the frequency of a

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Q1. write a program if a list contains a palindrome of e

- `l = [1,2,3,2,1]`

Q2. write a program to check whether 13 is a prime nu

Q3. Python program to convert the month name to a nu

Q4. Take value of length and breath from the user and

Q5. Take two numerical values from user and print gre

Q6. Calculate the percentage of class attend and say w
exam?

- `working_days = 280`

Q7. write a program to accept percentage from the use

- `< 40 -- Failed`
- `<55 & >=40 -- Fair`
- `<65 & >=55 -- Good`
- `greater or =65 -- Excellent`

Q8. Accept three sides of a triangle from user and che
scalene triangle?

- note :

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Q11. Write a program that takes employees salary as an input and calculates their bonus based on their years of service . The program should be as follows:

- If the years of experience is less than 5 : print(0)
- if the years of experience is more than 5 : calculate the bonus and print it credited.

Q12. write a program to accept two numbers from the user and calculate their sum using a function accordingly?

Q13. write a program to accept the three sides of a triangle and calculate its area.

Q14. create a program that calculates the library charge for a book borrowed and not returned. The charge structure is as follows:

- For exactly 5 days: Rs 2 per day
- For 6 to 10 days: Rs 3 per day
- For 11 to 15 days: Rs 4 per day
- For more than 15 days: Rs 5 per day

Q15. Write a program to check whether a person is eligible for a driving license or not?

- criteria for voting : more than or equal to 18 - eligible
- criteria for getting driving license: more than or equal to 21 - eligible

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- `l = [1, "abc", 1, "abc"]`

Q25. Write a program that prompts the user to enter the age of a person and whether they are:

- A child (0-12)
- A teenager (13-19)
- An adult (20-64)
- A senior (65 and above)

Q26. Write a program that asks the user to enter a day of the week: weekend (Saturday or Sunday) or a weekday?

Q27. Write a function that calculates the factorial of a number.

Q28. Create a program that takes a number and prints its square and cube.

Q29. create a program to accept the kilometers covered by a person and calculate the cost of the journey.

- First 10 km --- rs 11/km
- Next 90 km --- rs 10/km
- After that --- rs 9/km

Q30. Accept the market price from the user and calculate the amount to be paid after a discount of 20%.

- `amount to be paid = market_p - (market_p * discount)`
- `market_p > 10,000 --- discount = 20%`

